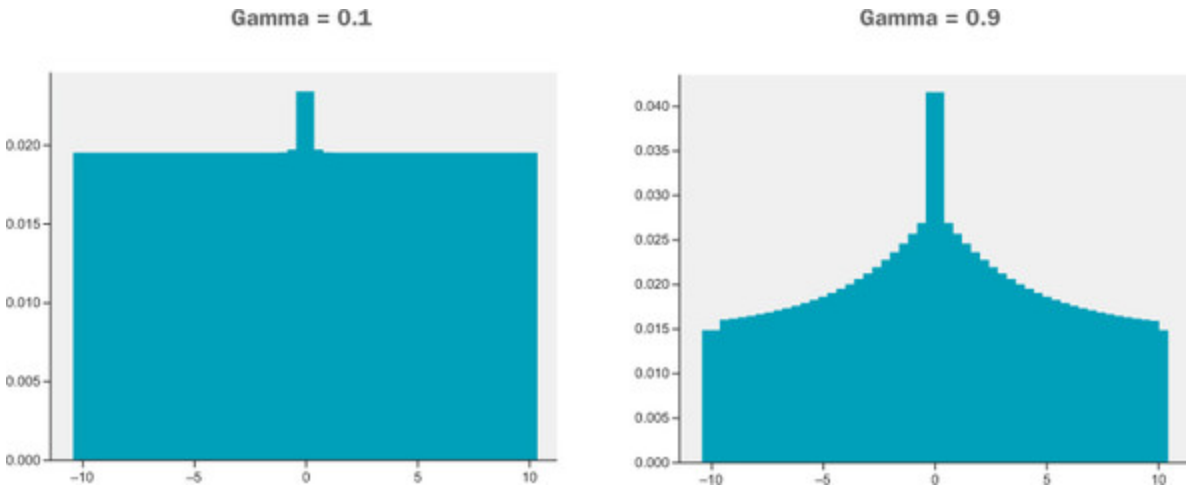
Figure 7.10. This figure shows how a uniform distribution changes with lower or higher values for gamma (the discount factor).



If we discount the future a lot, the posterior will be strongly centered around the recently observed reward. If we weakly discount the future, the observed reward will only mildly update the prior distribution, $Z(S_{t+1},A_{t+1})$. Since Freeway has sparse positive rewards in the beginning (because we need to take many actions before we observe our first win), we will set gamma so we only make small updates to the prior distribution.

In listing 7.1 we set up an initial uniform discrete probability distribution and show how to plot it.

Listing 7.1. Setting up a discrete probability distribution in numpy

```
import torch
import numpy as np
from matplotlib import pyplot as plt

vmin,vmax = -10.,10.                        1
nsup=51                                     2
support = np.linspace(vmin,vmax,nsup)       3
probs = np.ones(nsup)
probs /= probs.sum()
z3 = torch.from_numpy(probs).float()
plt.bar(support,probs)                      4
```

- ***1*** Sets the minimum and maximum values of the support of the distribution
- ***2*** Sets the number of elements of the support